MILITARY SPECIFICATION

COATS, FIREMEN'S, STRUCTURAL/ALUMINIZED PROXIMITY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for a flame resistant structural firefighters coat with a para-aramid and PBI blend outershell and a waterproof moisture barrier lining. In addition the coat has either a removable aluminized Para-aramid or a removable aluminized Para-aramid/PBI cover for proximity use.

1.2 Classification. The coat covered by this specification shall be of the following types, classes and sizes (see 6.2):

1.2.1 Types and classes.

Type I - Structural coat

Class 1 - Coat with neoprene coated lining
Class 2 - Coat with PTFE laminate lining

Type II - Coat cover (Aluminized)

1.2.2 Sizes.

Small	(34-36)	Large	(42-44)
Medium	(38-40)	Extra Large	(46-48)

2. APPLICABLE DOCUMENTS

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Officer in Charge, Navy Clothing and Textile Research Facility, 21 Strathmore Road, Natick, MA 01760-2490 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8415

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2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

FF-R-556	- Rivets, Solid, Small; Rivets, Split, Small; Rivets, Tubular, Small; and Caps, Rivets; General Purpose
FF-W-92	- Washers, Metal, Flat (Plain)
KK-L-2004	- Leather, Cattlehide, Deerskin and Horsehide, Chrome Tanned
NN-P-71	- Pallets, Material Handling, Wood,Stringer, Construction, 2 Way and 4 Way Entry (Partial)
QQ-N-290	- Nickel Plating (Electrodeposited)
QQ-P-416	- Plating, Cadmium (Electrodedeposited)
DDD-L-20	- Label; For Clothing, Equipage, and Tentage (General Use)

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MIL-B-371	- Braid, Textile, Tubular
MIL-C-3735	- Cuffs, Knit, Wrist and Ankle, and Cloth Knitted
MIL-P-15011	- Pallet, Material Handling, Wood Post Construction, 4-Way Entry
MIL-B-17757	- Boxes, Shipping, Fiberboard, (Modular Sizes)
MIL-C-19699	- Cloth, Coated (Nylon Taffeta)
MIL-E-20652	- Eyelet, Metallic, Rolled Flange Type, and Eyelet Washer
MIL-F-21840	- Fastener Tapes, Hook and Pile, Synthetic
MIL-C-24929	- Cloth, Laminated, Aluminized (Para-Aramid Base)
MIL-C-24939	- Cloth, Ripstop, Para-aramid/PBI, Water Repellent
MIL-C-24941	- Cloth, Laminated, Aluminized (Para-Aramid/PBI Base)
MIL-T-43636	- Thread, Aramid
MIL-C-43774	- Cloth, Plain or Pajama Check Weave, Aramid
MIL-T-44100	- Thread, Para-Aramid, Spun, Intermediate Modulus

STANDARDS

FEDERAL

FED-STD-191	- Textile Test Methods
FED-STD-311	- Leather, Methods of Sampling and Testing
FED-STD-751	- Stitches, Seams, and Stitchings

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MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	- Marking for Shippment and Storage
MIL-STD-147	- Palletized Unit Loads
MIL-STD-1666	- Classification of Defects for Evaluating Quality of Aluminized Firemen's Clothing
MS-27980	- Fastener, Snap, Style 2

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

LAWS AND REGULATIONS

U.S. POSTAL SERVICE MANUAL

(Copies of the manual may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Applications for copies should be addressed to the American Trucking Association, ATTN: Traffic Department, 1616 P Street, N.W., Washington, DC 20036).

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, Illinois 60606).

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JOINT COMMITTEE ON PRINTING (JCP) PUBLICATIONS

Government Paper Specification Standards

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Guide sample. Samples, when furnished, are solely for guidance and information to the contractor (see 6.3). Variations from this specification may appear in the sample in which case this specification shall govern.

3.2 First article. When specified, the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.3 and 6.2).

3.3 Materials.

3.3.1 Basic material (structural coat). The basic material for the structural coat shall be a water repellent para-aramid/PBI ripstop fabric conforming to MIL-C-24939.

3.3.2 Basic material (Aluminized cover). The basic material for the aluminized cover shall be either of the following fabrics. A highly reflective aluminum surface laminated to a knitted Para-Aramid base fabric conforming to Type II of MIL-C-24929 or a highly reflective aluminum surface laminated to a knitted para-aramid/PBI base fabric conforming to MIL-C-24941.

3.3.3 Lining (Class 1 coat). The material for lining the class 1 structural coat and for binding the facings of both classes 1 and 2 coats shall be a coated nylon taffeta conforming to Type II of MIL-C-19699. The binding shall be cut in strips of 1-1/8 ($\pm$ 1/8) inches wide and may be spliced.

3.3.4 Lining (Class 2 coat). The material for lining the class 2 structural coat shall be a two layer laminated cloth which is waterproof and moisture vapor permeable. The face of the fabric shall be a flame resistent, aramid pajama check weave cloth conforming to MIL-C-43774, color natural. The back side of the laminated cloth shall be a micropourous, expanded polytetraflourothylene film weighing 0.7 $\pm$ 0.3 oz/yd^2 as tested in 4.4.1 and an adhesive suitable for the purpose intended and compatible with the film. The face and back cloths shall be laminated together in such a manner that the assembled (finished) cloth shall comply with the requirements specified in Table I when tested as specified in 4.4.1 (see 6.5).

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TABLE I

Characteristic	Requirement
Moisture Vapor Transmission rate, Grams/m^2/24 hrs (min)	450
Hydrostatic Resistance, PSI (min)	50
Water Permeability After Synthetic Perspiration	No leakage

3.3.5 Thread. The thread for assembling the coat shall be a non-melting nylon (aramid), natural shade, conforming to Size B of MIL-T-43636. As an alternate, a thread conforming to Size T-50 of MIL-T-44100, natural in color may be used.

3.3.6 Leather. The leather for the shoulder protective panels, sleeve tabs, reinforceing snap hooks and for binding the sleeve edge shall be a flesh split cattlehide, natural color of tannage, of not less than 2-3/4 nor more than 3-1/2 ounce thickness conforming to type IV, class 2 of KK-L-2004 after silicone treatment specified in 3.3.6.1, except for the following;

a. Shrinkage temperature shall be not less than 105°C.
b. Tests for percent ash and chloroform-soluble material not required.

3.3.6.1 Silicone treatment. The leather shall be treated with an approved durable silicone compound (see 6.8) to lower its water water absorption. The treated leather shall not absorp more than 30 percent water when tested as specified in 4.4.1.

3.3.7 Cloth, corduroy. The material for the topcollar shall be 100% cotton corduroy. It shall have a minimum of eight wales per inch and weigh not less than 8.0 oz/yd^2. The cloth shall be fast dyed (no sulphur) either dark blue, dark green, black, chocolate brown or olive drab and have an approved durable flame retardant treatment (see 4.4.1 and 6.8). The dyed and treated cloth shall show good colorfastness to crocking and perspiration.

3.3.8 Hanger loop. The material for the hanger loop shall be cotton braid, plain, flat tubular, conforming to Type VII, class 1 of MIL-B-371, except that the requirements for colorfastness shall not apply. The color of the braid shall be black or a color to closely match the color of the collar.

3.3.9 Reflective tape. The reflective tape shall be in accordance with 3M style 8487 flourescent lime/yellow, 3 inch wide or equal (see 6.6 and 6.9).

3.3.10 Hook and pile fastener tape. The hook and pile fastener tapes shall conform to Type II, Class 1 of MIL-F-21840. The tape shall be 1 inch and 2 inch wide, as specified in Tables II and III. The color shall be natural, white or black.

3.3.11 Wristlet. The material for the wristlet shall be in accordance with Type IV, class 2 of MIL-C-3735.

3.3.12 Hardwear.

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3.3.12.1 Eyelets and washers. The eyelets and washers for the pocket drain holes shall conform to rolled flange type, M20652/1-BBE114 and M20652/1-BBW101 of MIL-E-20652.

3.3.12.2 Snap hooks and dee rings. The snap hooks and dee rings for fastening the front closure of the coat shall be cast malleable iron, brass, aluminum, or zinc conforming to the style and measurement requirements specified in Figure 3 (see 4.4.1). The finish for the malleable iron shall be cadmium plate conforming to Type I, Class 3, of QQ-P-416. The finish for the brass shall be nickel plate conforming to Class 2, 0.0005 inch minimum thickness of QQ-N-290.

3.3.12.3 Rivets, tubular with caps or washers. The rivets for attaching snaps and dee rings shall conform to type XII, class 3, grade B or F of FF-R-556, except that they shall be drilled to the head or partially drilled to the head and the the inside of the rivet shank shall be chamfered. The rivets shall be 9/64 inch nominal diameter and their length shall be sufficient to assure proper clinching. The caps shall conform to Type II of FF-R-556. The finish for the brass rivets and caps shall be nickel plate conforming to class 2 of QQ-N-290, except that the minimum thickness shall be 0.0005 inch. An aluminum washer conforming to type A, grade I, class C, No. 20 of FF-W-92 may be used in lieu of caps for securing the tubular rivets. The diameter of the aluminum washer shall be 0.170 to 0.180 inch for the inside and 0.470 to 0.500 inch for the outside.

3.3.12.4 Fasteners, snap. The snap fasteners for attaching the snap-in liner and for attaching the aluminized cover shall consist of a button and socket (female half) and a stud and eyelet (male half). The fasteners shall be style 2, 24-line (size 1), nickel plate or bright chrome finish conforming to MS-27980 and the following part numbers:

Button - 1N or 1C	Socket - 6N or 6C
Stud - 7N or 7C	Eyelet - 8N or 8C

3.3.13 Labels.

3.3.13.1 Label, Structural coat. Each coat shall have a combination size, instruction and identification label conforming to Type VI, Class 14 of DDD-L-20, except that the colorfastness requirements shall not apply (see 6.11). The label shall bear the following inscription:

COAT, FIREMEN'S, STRUCTURAL (CLASS 1) (EXAMPLE)
FIBER CONTENT:
  Outershell - 60% Para-aramid/40% PBI
  Lining - Nylon taffeta (EXAMPLE)
CONTRACT NO: DLA-100-00-0-0000 (EXAMPLE)
STOCK NO: 8415-00-000-0000 (EXAMPLE)
SIZE: MEDIUM (EXAMPLE)
NAME OF CONTRACTOR:
NAME OF MANUFACTURER: (If other than contractor)

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LAUNDERING INSTRUCTIONS
HAND WASH IN LUKE WARM WATER
DO NOT BLEACH
HANG TO DRY
DO NOT MACHINE WASH OR DRY CLEAN
WARNING

COAT MUST BE WORN WITH LINER INTACT AND ALL CLOSURES FASTENED WHEN IN USE. DO NOT KEEP IN DIRECT CONTACT WITH FLAME. DO NOT STORE THIS GARMENT IN DIRECT SUNLIGHT. KEEP GARMENT CLEAN - SOILING WILL REDUCE PROTECTIVE QUALITIES. FAILURE TO COMPLY WITH THESE INSTRUCTIONS MAY RESULT IN SERIOUS INJURY OR DEATH.

DO NOT REMOVE THIS LABEL

3.3.13.2 Warning label, structural coat. Each coat shall have a warning label conforming to Type IV, Class 9 of DDD-L-20 applied to the inside back lining (see 6.11). The label shall be applied to the lining by a red marking medium. The label shall be printed with block sansserif type with letters 3/4 to 1 inch in height. The label shall contain the following inscription:

WARNING
MUST BE WORN WITH
LINER IN PLACE

3.3.13.3 Label, proximity cover. Each proximity cover shall have a combination size, instruction and identification label conforming to Type VI, Class 14 of DDD-L-20 except that the colorfastness requirements shall not apply. The label shall bear the following inscription:

COVER, FIREMEN'S COAT(PROXIMITY)
FIBER CONTENT - 100% Para-Aramid (OR) 60% Para-Aramid/40% PBI
CONTRACT NO: DLA-100-00-0-0000 (EXAMPLE)
STOCK NO: 8415-00-000-0000 (EXAMPLE)
SIZE: MEDIUM (EXAMPLE)
NAME OF CONTRACTOR:
NAME OF MANUFACTURER: (If other than contractor)

INSTRUCTIONS

1. CAUTION: MUST BE USED ONLY AS AN COVER WITH STRUCTURAL COAT.
2. Hang to dry and store properly to prevent damage to aluminized surfaces.

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3. Clean by gently rubbing surface with cloth using mild soap, PERCHLORETHYLENE or ISOPROPANOL. When cleaning with perchlorethylene and isopropanol, ensure that you have adequate ventilation and that you use proper glovees and eye protection.
4. To maintain reflective properties and to extend service life, use proximity shell for intended purpose only.
5. Do not use aluminized paint to repair worn out surface as it will not provide protection.

Note: Aluminized coating is damaged by contact with fire extinguishing agent Potassium Bicarbonate Powder (PKP). Before storing garment with PKP residue present, the following is recommended.

1. Dry powder should be removed with dry brush or blower.
2. Wet powder should be removed by thoroughly flushing with water and wiping the surface down with a clean soft cloth. Damp or wet garments should be hung to dry.

3.3.13.4 Warning label, proximity cover. Each proximity cover shall have a warning label conforming to Type IV, Class 9 of DDD-L-20 applied to the inside back lining. The label shall be applied to the lining by a red marking medium. The label shall be printed with block sansserif type with letters 3/4 to 1 inch in height. The label shall contain the following inscription:

WARNING
MUST BE WORN WITH
STRUCTURAL COAT

3.3.14 Repair of aluminum film. The aluminum film used for patches for repairing uncoated or cracked areas on the finished cover shall be the same as specified for coating the base cloth. The area to be repaired shall not exceed 1 inch in the longest direction. The patch shall be completely bonded onto the aluminum film of the outer shell and shall be so placed that it extends not less than 1/2 inch beyond all edges of the uncoated or cracked area. No outer shell shall contain more than 3 patches. The unsupported aluminum film and self-curing adhesive used for repairing shall be obtained from the basic material supplier (see 3.3.2). The size of the patch shall not exceed 2 inches in any direction.

3.3.15 Instruction Manual. An instruction manual shall be furnished by the contractor with each aluminized cover and structural coat. The manual shall be inserted into the polyethylene bag during packaging, and positioned so that it is visible (see 5.1.1.1 and 5.1.1.2). The printing, collation, folding and stitching of the manual shall be in accordance with the standard sample (see 6.3). The cover shall be antique white book paper conforming to Joint Committee Printing-A-100, 140 pound basis weight 25 X 38 inches per 1000 sheets. The text paper shall be white offset book paper conforming to JCP-A-60, 120 pound basis weight 25 X 38 inches per 1000 sheets. A sample of the manual plus four glossy pictures shall be obtained from the contracting officer.

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3.4 Design. The coat shall be of a two piece design, an inner structural coat fabricated from a para-aramid/PBI fabric and lined with a moisture proof barrier fabric and an outer proximity cover fabricated from a highly reflective aluminized coated para-aramid/PBI blend fabric. The structural coat shall have a corduroy topcollar, sleeves with leather bound edges and side bellows type patch pockets with flaps. The front closure shall consist of protective flaps secured by means of three snap hooks and dee rings. Securing panels on both sides of the front opening are for attachment of the outer shell. The cover is designed to fit over the structural coat with openings and flaps made for the pockets. Attachment is made by snap fasteners and securing panels on both sides of the front opening.

3.4.1 Figures. Figures are furnished for informational purposes only. When inconsistencies exist between the written specification and the figures, the written specification shall govern.

3.5 Patterns. Standard patterns to be used to cut working patterns will be furnished by the Government (see 6.3). The patterns allow a 1/2 inch seam allowance for major joining seams, except as otherwise indicated in Tables II and III. The working patterns shall be identical to the Government patterns. Neither the Government patterns nor the working patterns shall be altered in any way except that additional notches for use during construction are allowed on the working patterns.

3.5.1 Pattern parts. The component parts of the coat shall be cut from the materials specified and in accordance with the number of parts required for the manufacturing process as follows:

MATERIAL	NOMENCLATURE OF PATTERN PARTS	CUT PARTS
	STRUCTURAL COAT (TYPE I)	
Base fabric	Front left side	1
	Front right side	1
	Back	1
	Sleeve	2
	Front facing left side	1
	Front facing right side	1
	Under collar	1
	Under collar stand	1
	Patch pocket (all sizes)	2
	Patch pocket flap (all sizes)	2
	Radio pocket (all sizes)	1
	Radio pocket flap (all sizes)	1
	Elbow patch	2
	Throat strap (all sizes)	2

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MATERIAL	NOMENCLATURE OF PATTERN PARTS	CUT PARTS
Aluminized	Securing panel left side	1
	Securing panel right side	1
Corduroy	Top collar	1
	Top collar stand	1
Moisture barrier	Front left side	1
	Front right side	1
	Back	1
	Sleeve	2
	Sleeve lining extension	2
	Wristlet stand	2
	Front facing left side	1
	Front facing right side	1
	Under collar	1
	Under collar stand	1
	Patch pocket reinforcement (all sizes)	2
	Patch pocket flap reinforcement (all sizes)	2
	Radio pocket reinforcement (all sizes)	1
	Radio pocket flap reinforcement (all sizes)	1
Leather	Protective shoulder	2
Templates	Left side marker for snap hook and fastener tape location	-
	Right side marker for dee rings and fastener tape location	-
	Right side for fastener tape location	-
	ALUMINIZED COAT COVER (TYPE II)	
Aluminized	Front left side	1
	Front right side	1
	Back	1
	Sleeve	2
	Radio pocket flap (all sizes)	1
	Radio pocket facing (all sizes)	2
	Pocket flap (all sizes)	2
	Pocket facing (all sizes)	4
	Neck facing	2
Leather	Buttonhole reinforcement	3
Template	Marker for buttonhole reinforcement on aluminized cover	-

3.6 Construction.

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3.6.1 Stitches, seams and stitchings. Stitch, seam and stitching types specified in Tables II and III shall conform to FED-STD-751. Whenever two or more methods for seam or stitch types are given for the same part of the operation any one may be used. Seam allowance shall be maintained with seams sewn so that no raw edges, runoffs, twists, pleats, puckers or open seams result. All seams shall start and finish evenly. Thread tension shall be maintained so that there is no tight or loose stitching. Where stitch type 401 is used, the looper (underthread) shall be on the inside of the garment. Top stitching and edge stitching shall be 1/16 to 1/8 inch from the edge, hook and pile fastener tape and reflective tape stitching shall be 1/16 to 1/8 inch from the tape edge. The bight of the overedge stitching shall be 3/16 to 1/4 inch. A knife shall be used on the overedge machine with the guide set so thaty only the ravelled edges are trimmed. In order to minimize the size of the needle hole in the aluminized fabric, the needle used in the seaming operations shall be the smallest size which will accommadate the specified size thread.

3.6.2 Thread breaks and end of seams. End of all seams and stitching when not caught in other seams or stitching shall be backtacked not less than 3/8 inch. The ends of a continuous line of stitching shall be overlapped not less than 1/2 inch. Thread breaks (all stitch type) shall be repaired by stitching back of the break not less than 1/2 inch. Skipped stitches or thread breaks on 401 stitch type may be repaired by using 301 stitch type. When ends of overedge stitching are not caught in other seams or stitching, the ends shall be tacked. When stitch type 401 is permitted as an option, it shall not be used unless the ends are caught in other seams of stitching.

3.6.3 Stitches per inch. The minimum and maximum number of stitches per inch shall be as specified in Tables II and III.

3.7 Manufacturing operations requirements. The coat shall be manufactured in accordance with the operation requirements specified in Tables II and III. The supplier is not required to follow the exact sequence of operations listed provided that the finish coats are identical to those produced by following the sequence of operations as listed in Tables II and III.

3.7.1 Size marking. The component parts of the coat shall be bundled, marked or ticketed to assure proper size assembly. The use of metal fastening devices or sewn-on tickets for marking is prohibited. Adhesive tickets, when used, shall not discolor the material and the adhesive mass shall not adhere to the material upon removal of the tickets.

NOTE: The use of ink pad numbering machines, rubber stamp or pencil is allowed, provided the numbering does not show on the outside of the coat and wherever possible, is covered by the seam allowance.

3.7.2 Abbreviations in table of operations. The abbeviations used in tables II and III are as follows:

Stch	-	Stitch
in	-	inch
Ndl	-	Needle
Bob	-	Bobbin
Lpr	-	Looper

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TABLE II - CONSTRUCTION OF STRUCTURAL COAT

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	NDL BOB/ LPR

1. Cutting.

a. Cut the structural coat in strict accordance with patterns which show directional lines, marks and notches for proper assembly. Directional lines shall not appear on the fabrics. Drill holes are not permitted.

b. The component parts of the base fabric may be cut from ends and when so cut shall approximate the color of the major cut parts.

c. The component parts of the lining may be cut from ends.

d. Cut the leather sleeve tabs 1 inch wide by 5-1/2 to 6 inches in length.

e. Cut the wristlets to finish 4 (± 1/4) inch long on the finished coat.

f. Cut 2 inch wide leather strips to bind the edges of the sleeve openings.

g. The face side of the materials to be assembled shall be considered as follows:

Outershell - as marked
Lining - uncoated side
Corduroy - cord side

NOTE: During the manufacturing of this garment, the water repellent coated side of the lining shall face away from the wearer in the finished coat.

2. Replacement of defective components.

During the spreading, cutting and manufacturing operations, components having material defects or damages that are classified as damages in MIL-STD-1666 and Section 4 shall be removed and replaced with nondefective and properly matched components.

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TABLE II - CONSTRUCTION OF STRUCTURAL COAT

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	NDL	BOB/ LPR
3.	Seam direction.					
	The direction of the joining seams shall be lapped as follows: 1. Back over fronts at sides and shoulders. 2. Front and back body over sleeves at armholes. 3. Back portion of sleeve over front portion of sleeve at underarm.					
4.	Make securing panels.					
	a. Fold securing panel for the left front in half, back to back with the long side edges meeting. Turn top, bottom and side raw edges in 1/8 to 3/16 inch and edgestitch all edges.	301	SSc-1	6-8	B	B
	b. Center a strip of pile fastener tape down the length of the left front securing panel and stitch along all edges.	301	LSbj-1	6-8	B	B
	c. Fold securing panel for the right front in half, back to back, with the long edges meeting. Turn top, bottom and side edges in 1/8 to 3/16 inch and edgestitch all edges.	301	SSc-1	6-8	B	B
	d. Fold right front securing panel in half again with the long edges meeting, stitch the length of the panel 1/8 (± 1/16) inch from the folded edge.	301	OSf-1	6-8	B	B
	e. Open up right front securing panel and position three, 1-1/2 inch by 2 inch hook fastener tape pieces on the left side in accordance with the pattern marker. Stitch along all edges of the fastener tape.	301	LSbj-1	6-8	B	B
	f. Position three D rings to the underside of the securing panel in accordance with the pattern marker. Securely rivet the three D rings to the panel.					
	g. Position a strip of pile fastener tape down the length of the right side of the opened right front securing panel with outside edge of the tape 3/8 (± 1/16) inch from the folded edge, stitch along all edges of the fastener tape.	301	LSbj-1	6-8	B	B

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TABLE II - CONSTRUCTION OF STRUCTURAL COAT

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	NDL	BOB/ LPR
5.	Attach elbow patches and reflective tape to outer (PBI) sleeves.					
	a. Turn sides and front edges of elbow patch in 3/8 inch and double stitch 1/4 to 5/16 inch gauge to sleeve, as indicated by notches on sleeve, with the outer row of stitching 1/8 to 3/16 inch from the folded edge. Top and bottom edges of the elbow patch shall be parallel to the bottom edge of the sleeve.	301 or 401	LSd-2	6-8	B	B
	b. Position bottom edge of reflective tape in accordance with marks on sleeve pattern and extend tape from sleeve edge to sleeve edge. Stitch each side of the tape to the sleeve.	301 or 401	LSbj-1	6-8	B	B
6.	Join shoulders.					
	a. Join, face to face, 3/8 to 1/2 inch from raw edge, the fronts and back at shoulder. Turn back panel over seam allowances and topstitch 1/8 to 3/16 inch from folded edge.	301 or 401	LSq-2	6-8	B	B
7.	Attach protective shoulder piece and reflective tape.					
	a. Position the protective shoulder piece on the shoulder matching the mark with the shoulder seam and as indicated on the patterns. The armhole edge of the protective shoulder piece shall finish 3/4 (+ 1/16) inch from the raw armhole edge. Double stitch the protective shoulder piece to the coat, 1/4 to 5/16 inch gauge, with the outer row of stitching 1/8 to 3/16 inch from the raw edge.	301	LSbj-2	6-8	B	B
	b. Position reflective tape on the upper and lower fronts of the coat, with the bottom edge of the tape in accordance with the notches on the patterns. The tape shall extend the distance indicated on the patterns, be parallel to the bottom edge of the coat and the lower tape section shall be covered by the patch pocket. Stitch each side of the reflective tape to the coat.	401	LSbj-1	6-8	B	B

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TABLE II - CONSTRUCTION OF STRUCTURAL COAT

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	NDL	BOB/ LPR
7.	Attach protective shoulder piece and reflective tape. (Cont'd)					
	c. Position reflective tape on the upper and lower back of the coat, with bottom edge of the tape in accordance with notches on the patterns. The tape shall be parallel to the bottom edge of the coat and extend across the entire back width. Stitch each side of the reflective tape to the coat.	401	LSbj-1	6-8	B	B
8.	Assemble and attach patch pockets.					
	a. Position reflective tape on the patch pockets as indicated by pattern marks and stitch each side and both ends of the tape to the pocket.	301	LSbj-1	6-8	B	B
	b. Position patch pocket reinforcement behind the patch pocket outershell. The two plies may be stay stitched around the outer edges.	301 or 401	SSa-1	6-8	B	B
	c. Position three metal eyelets on each patch pocket in accordance with marks on pattern and attach with the center of each eyelet 7/8 (±1/8) inch from the finished bottom edge of pocket.					
	d. Turn under the top raw edge of pocket 3/8 to 1/2 inch and fold top as indicated by pattern marks forming a 3 inch hem. Stitch 1/8 to 3/16 inch and again 2-5/8 to 2-3/4 inch from the top folded edge.	301	EFt-2	6-8	B	B
	e. Position a 2 inch by 3 inch piece of hook fastener tape on the patch pocket as indicated by the marks on the patterns. Stitch along all edges.	301	LSbj-1	6-8	B	B
	f. Fold edge of patch pocket on fold line closest to the raw edge, face to face and stitch 1/8 to 3/16 inch from folded edge.	301	OSf-1	6-8	B	B
	g. Fold edge of patch pocket on second fold line, back to back and stitch 1/8 to 3/16 inch from folded edge.	301	OSf-1	6-8	B	B

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TABLE II - CONSTRUCTION OF STRUCTURAL COAT

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	NDL	BOB/ LPR
8.	Assemble and attach patch pockets. (Cont'd)					
	h. Turn raw pleated front side edge of patch pocket under 3/8 to 1/2 inch. Lay patch pocket flat with pleats closed. Turn bottom raw edge under 3/8 to 1/2 inch and position on coat front as indicated by pattern marks, with back raw edge of pocket and back raw edge of front even. Topstitch front side and bottom edge of pocket to coat front, stitching shall be 1/8 to 3/16 inch from front and bottom folded edge of patch pocket.	301	LSd-1	6-8	B	B
	i. The finished patch pocket for all sizes shall measure 10 1/2 (± 3/8) inches deep at center of pocket. The width of the pocket shall be 8-3/4 (± 3/8) inches when measured across top edge of pocket from turned edge of bellows to folded edge of side seam.					
	NOTE: Care shall be exercised so as not to catch the outer pleated edge of pocket in the stitching of the pocket front side edge.					
9.	Make and attach patch pocket flaps.					
	a. Position a 1 inch by 3 inch piece of pile tape on the underside of the patch pocket flap as indicated by marks on patterns. Stitch along all edges.	301	LSbj-1	6-8	B	B
	b. Position the patch pocket reinforcement on the inside of the top layer of the patch pocket outershell and edge stitch together along both side edges.	301	SSa-1	6-8	B	B
	c. Fold patch pocket flap in half face to face, as indicated by marks on patterns and stitch 3/8 inch to 1/2 inch from the raw side edges.	301	SSe-2(a)	6-8	B	B
	d. Trim points, turn patch pocket flap to finished position and completely force out corners. Stitch the turned in sides and bottom folded edge 1/8 inch to 3/16 inch gauge.	301	SSe-2(b)	6-8	B	B

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TABLE II - CONSTRUCTION OF STRUCTURAL COAT

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	NDL	BOB/ LPR
9.	Make and attach patch pocket flaps. (Cont'd)					
	e. Turn under 1/4 to 3/8 inch the top raw edge of the patch pocket flap and double stitch 1/4 to 5/16 inch guage to each front as indicated by marks on patterns. The inner row of topstitching shall be 1/4 (± 1/8) inch above top folded edge of the pocket.	301 or 401	LSbl-2	6-8	B	B
	or					
	f. Position the patch pocket flap face down on coat front, with raw edges facing towards the bottom of the coat and 3/8 inch below the mark on the pattern. Stitch 1/4 to 5/16 inch from raw edge. Turn flap down to finished position and topstitch 3/8 to 7/16 inch from top folded edge of flap.	301 or 401	LSbl-2	6-8	B	B
	NOTE: Back edge of the flap shall be free of side seams.					
	g. Finished patch pocket flaps for all sizes shall measure 8-1/2 (± 1/4) inches long when measured along the bottom edge and 3-3/4 (± 1/4) inches wide at side and center.					
	h. Stitch a 5/8 inch vertical bartack at the top corners of the patch pockets, a 5/8 inch horizontal bartack at the bottom front corner of the patch pocket and a 5/8 inch horizontal bartack at each top corner of the patch pocket flap. The bartacks shall be superimposed on the previous line of stitching.	Bartack		42 per bartack	B	B
10.	Attach sleeves and close side seams.					
	a. Join face to face 3/8 to 1/2 inch from the raw edge, the sleeves to the armhole, matching notches. Turn back the front and back panels over seam allowances and topstitch 1/8 to 3/16 inch from folded edge.	301 or 401	LSq-2	6-8	B	B

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TABLE II - CONSTRUCTION OF STRUCTURAL COAT

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	NDL	BOB/ LPR
10.	Attach sleeves and close side seams. (Cont'd)					
	b. Join face to face 3/8 to 1/2 inch from the raw edge, the side of coat and underarm sleeve, simultaneously catching the back edge of the patch pocket and ends of the reflective tape in one continous operation. Turn back the back panel and sleeve over the seam allowance and topstitch 3/16 to 1/4 inch from folded edge.	301 or 401	LSq-2	6-8	B	B
	c. As an alternate, operations 10.a. and 10.b. may be performed using seam type LSb-2.					
11.	Make radio pocket and flap and attach to right front.					
	a. Position radio pocket reinforcement behind the radio pocket outershell. The two plies may be stay stitched around the outer edge.	301 or 401	SSa-1	6-8	B	B
	b. Turn under the top raw edge of pocket 1/4 inch and fold top again as indicated by pattern marks forming a 1-1/4 inch hem. Stitch 1/8 to 3/16 inch and again 1 to 1-1/8 inch from the folded edge.	301	EFt-2	6-8	B	B
	c. Position and attach two metal eyelets on the bottom end of the radio pocket as indicated by pattern marks.					
	d. Fold radio pocket according to fold lines. Position side pocket edges face to face and stitch 3/8 (± 1/16) inch from raw edges.	301	SSa-1	6-8	B	B
	e. Turn pocket to finished position and edge stitch 1/16 to 1/8 inch along all fold lines.	301	EFa-1	6-8	B	B
	f. Center a 2 inch by 3 inch piece of hook fastener tape to the pocket front as indicated by pattern marks and stitch along all edges.	301	LSbj-1	6-8	B	B
	g. Position the radio pocket flap reinforcement on the top layer of the radio pocket flap outershell and edge stitch all edges.	301	SSa-1	6-8	B	B

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TABLE II - CONSTRUCTION OF STRUCTURAL COAT

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	NDL	BOB/ LPR
11.	Make radio pocket and flap and attach to right front. (Cont'd)					
	h. Fold radio pocket flap face to face, and stitch 3/8 (+ 1/16) inch from side raw edges. Trim corners and turn to finished position and edge stitch turned edges.	301 and 301	SSe-2(a) SSe-2(b)	6-8 6-8	B B	B B
	i. Center a 1 inch by 3 inch piece of pile fasterner tape on the underside of the finished end of the radio pocket flap in accordance with marks on patterns and stitch along all edges.	301	LSbj-1	6-8	B	B
	j. Turn under raw edges of radio pocket 1/4 inch, position on right front in accordance with marks on patterns and edge stitch to coat.	301	LSd-1	6-8	B	B
	k. Turn under 1/4 inch to 3/8 inch the top raw edge of the radio pocket flap and double stitch 1/4 to 5/16 inch guage to right front as indicated by marks on patterns. The inner row of top stitching shall be 3/8 (+ 1/8) inch above top folded edge of the pocket.	301 or 401	LSba-2	6-8	B	B
	or					
	l. As an alternate, position the radio pocket flap face down on right front, with raw edges facing towards the bottom of the coat and 3/8 inch below the marks on the patterns. Stitch 1/4 to 5/16 inch from the raw edge. Turn flap down to finished position and topstitch 3/8 to 7/16 inch from top edge of flap.	301 or 401	LSbl-2	6-8	B	B
	m. Stitch a 5/8 inch vertical bartack at the top corners of the pocket and a 5/8 inch horizontal bartack at the top corners of the pocket flap. Bartacks shall be superimposed on the previous lines of stitching.	Bartack		42 per bartack	B	B
	n. The front of the finished radio pocket shall measure 8-1/2 (+ 1/4) inches deep and 4 (+ 1/4) inches wide. The flap shall measure 4 (+ 1/4) inches wide by 4-1/2 (+1/4) inches long.					

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TABLE II - CONSTRUCTION OF STRUCTURAL COAT

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	NDL	BOB/ LPR
12.	Assemble facings.					
	a. Position and edgestitch a combination label (see 3.3.13.1) centered on the right facing, to finish 6 to 8 inches below top edge of facing on finished coat.	301	LSbj-1	6-8	B	B
	b. Superimpose the lining facings on the outershell facing and edge stitch along all raw edges.	301	SSa-1	6-8	B	B
	c. Position facing assemblies face to face, and stitch 3/8 (± 1/16) from raw edge. Spread seam open and flat.	301	SSa-1	6-8	B	B
	d. Bind inside edge of joined facings with a 1-1/8 (± 1/16) inch wide strip of coated nylon taffeta. The raw edges of the facing shall extend within and to the foldline of the binding.	301	BSa-1	6-8	B	B
	e. Position and attach snap fastener sockets and caps to finished facing as marked on pattern. Socket shall face away from the wearer.					
13.	Make collar.					
	a. The assembled collar shall consist of an outershell undercollar, a vapor barrier interlining and a corduroy topcollar. The collar assembly shall be joined to an collarstand assembly which consists of the same three layer configuration with a hanger loop attached.					
	b. Assemble the top and bottom plies of the throat strap face to face. Stitch 3/8 (± 1/16) inch from the sides and pointed end.	301	SSe-2(a)	6-8	B	B
	c. Turn throat strap to the finished position and edge stitch along the sides and pointed end.	301	SSe-2(b)	6-8	B	B
	d. Position a 1-1/4 inch by 2 inch piece of pile fastener tape on each side of the throat strap as indicated by the marks on the patterns. Stitch tapes along all edges.	301	LSbj-1	6-8	B	B

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TABLE II - CONSTRUCTION OF STRUCTURAL COAT

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	NDL	BOB/ LPR
13.	Make collar. (Cont'd)					
	e. Position throat strap on undercollar with the raw edge in accordance with the marks on the pattern, and the pointed end of the strap lying across the undercollar with the point facing the opposite end of the collar. Stitch 1/4 to 5/16 inch from the raw edge . Fold strap over itself and topstitch 3/8 to 7/16 inch from folded edge.	301 or 401	LSbk-2 or LSbl-2	6-8	B	B
	f. Position a 1-1/4 by 2 inch piece of hook fastener tape to each end of the undercollar as indicated by marks on patterns. Stitch tapes along all edges.	301	LSbj-1	6-8	B	B
	g. Superimpose the undercollar vapor barrier on the under collar and edge stitch along all raw edges.	301	SSa-1	6-8	B	B
	h. Join face to face, 5/16 to 3/8 inch, the undercollar assembly and the topcollar along the outer edges.	301 or 401	SSe-2(a)	6-8	B	B
	i. Trim corners and turn collar to finished position, working out edges. Topstitch outer edges 1/4 to 5/16 inch from folded edge.	301 or 401	SSe-2(b)	6-8	B	B
	j. Superimpose the undercollar stand vapor barrier on the under collar stand and edge stitch along all raw edges.	301	SSa-1	6-8	B	B
	k. Match the notch on the short side of the undercollar stand assembly, face to face, with the notch on the undercollar. Stitch 5/16 to 3/8 inch from the raw edge. Turn the undercollar stand over the seam allowance and topstitch 1/16 inch from the folded edge.	301	LSq-2	6-8	B	B

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TABLE II - CONSTRUCTION OF STRUCTURAL COAT

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	NDL	BOB/ LPR
13.	Make collar. (Cont'd)					
	1. Match the notch on the short side of the topcollar stand face to face, with the notch on the topcollar. Stitch 5/16 to 3/8 inch from the raw edge. Turn the topcollar stand over the seam allowance and topstitch 1/16 inch from the folded edge. The bottom raw edges may also be stitched together for ease in attaching collar assembly to the neckline of the coat.	301	LSq-2	6-8	B	B
	m. A 5 inch length of hanger loop tape shall be pre-stitched to the bottom raw edge of the the neck portion of collar stand.	301 or 401	SSab	6-8	B	B
	n. When attached to the finished collarstand, the hanger loop shall be centered between the ends of the collar and its opening shall measure 3 (± 1/4) inches long between attached ends.					
14.	Prepare lining assembly.					
	a. Join 3/8 to 1/2 inch from raw edge the shoulder seam, side seam, sleeve inseam and armhole seam. The side seam and sleeve inseam shall be joined in one continous operation.	301 or 401	LSq-2(a)	6-8	B	B
	b. Turn one ply back over seam allowance and topstitch 1/8 to 3/16 inch from folded edge.	301 or 401	LSq-2(b)	6-8	B	B
	c. As an alternate, operation a and b may be performed in one operation by joining with a double lapped and double stitched seam, 1/4 inch gauge.	301 or 401	LSc-2	6-8	B	B
	d. Apply warning label, (see 3.3.13.2), horizontally across back of lining so that the top edge of label will finish 6 to 8 inches below collar seam on the finished coat.					

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TABLE II - CONSTRUCTION OF STRUCTURAL COAT

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	NDL	BOB/ LPR
14.	Prepare lining assembly. (Cont'd)					
	e. Position narrow ends of lining wristlet stand face to face and stitch 1/8 to 3/16 inch from raw edge.	301 or 401	SSa-1	6-8	B	B
	f. Fold wristlet stand back to back in accordance with the fold line on the pattern and stay stitch the periphery together 1/8 to 3/16 inch from the raw edge.	301 or 401	SSa-1	6-8	B	B
	g. Position narrow ends of sleeve lining extension face to face and overedge stitch 1/4 inch from raw edge.	515 or 516 or 519	SSa-1	6-8	B	B
	h. Position wristlet stand inside the knit wristlet with raw edges of wristlet and raw edgse of wristlet stand even. Stitch 1/8 to 3/16 inch from edge. The cuff shall be stretched so that the wristlet stand is flat and smooth.	301	SSa-1	6-8	B	B
	i. Position sleeve lining extension outside of the wristlet with raw edge even with the raw edges of wristlet. Stitch 1/8 to 3/16 inch from edges with the cuff stretched so that the extension is flat and smooth. The joining seams of operations e and g shall match up.	301	SSa-1	6-8	B	B
	j. Position turned edge of wristlet stand face to face with bottom of sleeve lining and stitch 1/8 to 3/16 inch from raw edges, simultaneously inserting and catching the leather sleeve tabs one at the underarm seam and the other directly opposite. For ease in assembly, leather sleeve tabs may be pre-stitched to bottom of sleeve lining before attachment of wristlet stand.	301	SSa-1	6-8	B	B
	k. Position assembled collar on the neckline of the assembled lining according to notches, face to face, and stitch 1/4 (± 1/16) inch from the raw edge, simultaneously catching the hanger loop.	301	SSa-1	6-8	B	B

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TABLE II - CONSTRUCTION OF STRUCTURAL COAT

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	NDL	BOB/ LPR
14.	Prepare lining assembly. (Cont'd)					
	l. Position pre-assembled facing on assembled lining with fronts and neckline edges even, and with the facing lining and the coat lining face to face. Stitch 3/16 to 1/4 inch from raw edges, thru all plies.	301 or 401	SSa-1	6-8	B	B
	m. Stitch thru all plies the entire length of facing 2 (± 1/4) inches from folded edge of binding. Stitching shall terminate at the bottom edge of collar.	301 or 401	SSa-1	6-8	B	B
15.	Join prepared coat assemblies.					
	a. Position edges of outershell and lining assemblies even at neckline and fronts with the outershell side of the facing, face to face with the outershell.					
	b. Join, 1/4 inch from raw edge, front edges across steps, and along neckline in one continuous operation. Trim step corners, turn coat to finished position and work out step corners and front edges.	301 or 401	SSe-2(a) and SSq-2(a)	6-8	B	B
	c. Edgestitch, 1/8 to 3/16 inch from folded edge, the fronts and neckline of coat in one continuous operation.	301	SSe-2(b) and SSq-2(b)	6-8	B	B
16.	Hem bottom.					
	Turn under entire bottom edge of coat 1/4 to 3/8 inch and fold again to form a 1 (± 1/8) inch hem, as indicated by marks on pattern. Ends of hem may be folded square of diagnally. Stitch top edge of hem and diagnally folded ends 1/8 to 3/16 inch from folded edge. Square folded ends shall be stitched 1/4 to 5/16 inch from folded front edge.	301	EFb-1	6-8	B	B

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TABLE II - CONSTRUCTION OF STRUCTURAL COAT

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	NDL	BOB/ LPR
17.	Attach securing panels.					
	a. Position the right securing panel to the right front of coat, with the dee rings against the face of the outershell and the inside corner of the panel on the notch for the collar. The folded front edge of the panel shall be parallel with the front edge of the coat. Stitch panel to coat 1/8 to 3/16 inch from the back edge of the panel.	301	LSbj-1	6-8	B	B
	b. Position the joined edge of the left securing panel to the left front edge of the coat with the pile fastener tape positioned against the face of the outershell. Stitch panel to coat 1/8 to 3/16 inch from the front edge.	301	SSaa-1	6-8	B	B
18.	Stay stitch sleeve bottom.					
	Join the outershell and sleeve lining extension at bottom sleeve edge, 1/4 inch from raw edges.	301	SSa-1	6-8	B	B
19.	Bind sleeve bottoms.					
	a. Position a two inch wide leather strip onto the inside of sleeve opening approximately 7/8 inch from raw edge of sleeve opening. Stitch 1/8 to 3/16 inch from the edge through all plies of material, overlapping ends of the strip approximately 1 to 1-1/2 inch near the underarm seam. Turn strip to the outside of the sleeve opening and stitch 1/8 to 3/16 inch from the edge along free edge and across the overlapped portion through all plies of material.	301	LSa-1 (each edge)	6-8	B	B
	or					

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TABLE II - CONSTRUCTION OF STRUCTURAL COAT

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	NDL	BOB/ LPR
19.	Bind sleeve bottoms. (Cont'd)					
	b. As an option, fold leather strip in half lengthwise using a folder, stitch each sleeve opening with a single or double row of stitching 1/8 to 3/16 inch from edges of leather strip, through all plies of material, overlapping ends of strip approximately 1 to 1-1/2 inches near the underarm seam. When double row of stitching is elected, the gauge of stitching shall be 1/4 inch. The ends of the sleeves shall extend within and to the folded edge of the leather strips.	301	BSa-1 or BSa-2	6-8	B	B
20.	Finish off coats.					
	a. Position four 1 by 2 inch pieces of hook fastener tape on the right front of the cloth outershell as indicated by pattern marker. Stitch along all edges of fasterner tape.	301	LSbj-1	6-8	B	B
	b. Position four 1 inch by 2 inch pieces of hook fastener tape on the left front of the cloth outershell as indicated by pattern marker. Also, position four 1 inch by 2 inch pieces of pile fastener tape on the inside of the left front directly behind the previously positioned hook tape. Stitch along all edges of fasterner tape, catching both pieces of tape in the stitching.	301	LSbj-1	6-8	B	B
	or					
	c. As an option to operation b. the fastener tapes may be stitched to each side indivually.	301	LSbj-1	6-8	B	B
	d. Rivet three snaps hooks on the left front in accordance with the pattern marker. Rivets shall not catch the facing and shall be reinforced with a leather patch on the inside.					

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TABLE II - CONSTRUCTION OF STRUCTURAL COAT

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	NDL BOB/ LPR
20.	Finish off coats. (Cont'd)				
	e. Attach snap fastener studs, one to each shoulder seam centered on seam and positioned 1-1/4 inch from undercollar stand joining seam. Attach a snap fastener stud to back hem mid-way between side seams and centered 3/4 inch from finished bottom edge. Studs shall be on the outside of the coat and face away from the wearer. Care shall be taken so that shoulder snap fasteners do not catch the inside facing.				
	f. Attach snap fastener sockets and caps to the ends of the leather sleeve tabs. The snap fasteners shall be centered 1/2 inch from the ends of the tab and the sockets shall face away from the wrist as worn. This operation may be done prior to the attachment of the tabs to the end of the sleeve opening.				
21.	Clean coat.				
	a. Trim all thread ends. b. Remove all loose threads. c. Remove all spots and stains.				

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TABLE III - CONSTRUCTION OF ALUMINIZED COVER

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	NDL BOB/ LPR

1. Cutting.

a. Cut the cover in strict accordance with patterns which show directional lines, marks and notches for proper assembly. Directional lines shall not appear on the fabrics. Drill holes are not permitted.

b. The component parts of the aluminized fabric may be cut from ends and when so cut shall finish with the aluminized side face up.

c. Cut triangular leather pocket opening reinforcements to finish 2 inches on the base and 1-1/2 inches on the other two sides.

d. Cut 2 inch wide leather strips to bind the edges of the sleeve opening.

e. The face side of the cover material is the aluminized side.

2. Replacement of defective components.

During the spreading, cutting and manufacturing operations, components having material defects or damages that are classified as damages in MIL-STD-1666 shall be removed and replaced with nondefective and properly matched components.

3. Seam direction.

The direction of joining seams shall be lapped as follows:

1. Back over fronts at sides and shoulders.
2. Front and back body over sleeves at armhole.
3. Back portion of sleeve over front portion of sleeves at underarm.

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TABLE III - CONSTRUCTION OF ALUMINIZED COVER

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	NDL	BOB/ LPR
4.	Bind patch pockets openings.					
	a. Slash patch pocket openings as indicated by marks on patterns. Slash shall extend into the side seam and the front edge of the slash shall be tongue notched 1/4 inch.					
	b. Fold patch pocket facings, back to back, as indicated by marks on patterns and stitch 3/8 to 5/16 inch from the folded edges.	301	OSf-1	6-8	B	B
	c. Position facings with folded edges abutted and ends even and tack together across each end 1/8 inch from raw edges.	301	SSa-1	6-8	B	B
	d. Fold raw edges of patch pocket opening under 1/4 inch. Position the folded edges of the patch pocket facings under the top and bottom of the patch pocket openings, with approximately 3/8 inch of the facings showing. Stitch the width of the patch pocket opening 1/16 inch from the folded edge of the top and bottom of the patch pocket openings and across the front edge end thru all plies.	301	SSc-1	6-8	B	B
	e. Stitch 1 (± 1/16) inch down from the folded edge of the lower patch pocket opening continuing across the front edge end and ending the line of stitching 5/8 inch above the top pocket opening.	301	SSc-1	6-8	B	B
	f. Center a 1 inch by 2-1/2 inch piece of pile tape on the inside of the lower edge of the patch pocket opening in accordance with marks on patterns. Stitch tape along all edges.	301	LSbj-1	6-8	B	B
	g. Postition a 1 inch by 2-1/2 inch piece of hook tape on the outside of the lower patch pocket opening as indicated by marks on pattern. Stitch tape along all edges.	301	LSbj-1	6-8	B	B
	h. As an alternate, operations 4.f and 4.g may be done at the same time.	301	LSbj-1	6-8	B	B

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TABLE III - CONSTRUCTION OF ALUMINIZED COVER

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	NDL	BOB/ LPR
5.	Make and position patch pocket flaps.					
	a. Fold patch pocket flap in half face to face, as indicated by marks on patterns and stitch 3/8 inch to 1/2 inch from the raw side edges.	301	SSe-2(a)	6-8	B	B
	b. Turn patch pocket flap to finished position and stitch the turned in sides and bottom folded edge 1/4 inch to 5/16 inch gauge. The corners of the flap shall be completely forced out.	301	SSe-2(b)	6-8	B	B
	c. Position a 1 inch by 2-1/2 inch piece of pile tape on the underside of the patch pocket flap as indicated by marks on patterns. Stitch tape along all edges through all plies.	301	LSbj-1	6-8	B	B
	d. Turn under 1/4 inch to 3/8 inch the top raw edge of the patch pocket flap and double stitch 1/4 to 5/16 inch gauge to each front as indicated by marks on patterns. The upper row of topstitching shall be 1/8 (± 1/16) inch below top folded edge of the pocket.	301 or 401	LSba-2	6-8	B	B
	or					
	e. As an alternate, position the patch pocket flap face down on coat front, with raw edges facing towards the bottom of the coat and 3/8 inch below the marks on the patterns. Stitch 1/4 to 5/16 inch from raw edge. Turn flap down to finished position and topstitch 3/8 to 7/16 inch from top folded edge of flap.	301 or 401	LSbl-2	6-8	B	B
	NOTE: Back edge of the flap shall be free of side seams.					

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TABLE III - CONSTRUCTION OF ALUMINIZED COVER

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	NDL	BOB/ LPR
5.	Make and position patch pocket flaps. (Cont'd)					
	f. Finished patch pocket flaps for all sizes shall measure 8-1/4 (+ 1/4) inches long when measured along the bottom edge and 3 (+1/4) inches wide at sides and center.					
	g. Position a leather triangular shaped reinforcement piece across the front end of the patch pocket opening with the base of the triangle parallel with the front edge of the pocket flap. Stitch 1/16 to 1/8 inch from all raw edges of the triangle.	301	LSbj-1	6-8	B	B
6.	Bind radio pocket opening.					
	a. Slash radio pocket opening as indicated by marks on patterns. Ends of slash shall be tongue notched 1/4 inch.					
	b. Fold radio pocket facings, back to back as indicated by marks on patterns and stitch 3/8 to 5/16 inch from the folded edges.	301	OSf-1	6-8	B	B
	c. Position facings with folded edges abutted and ends even and tack together across each end, 1/8 inch from raw edges.	301	SSa-1	6-8	B	B
	d. Fold raw edges of radio pocket opening under 1/4 inch. Position the folded edges of the radio pocket facings under the top and bottom of the patch pocket openings, with approximately 3/8 inch of the facing showing. Stitch the width of the radio pocket opening 1/16 inch from the folded edge of the top and bottom of the radio pocket openings and across the ends thru all plies.	301	SSc-1	6-8	B	B
	e. Stitch 1 (+ 1/16) inch down from the folded edge across the lower radio pocket opening and across both ends to 5/8 (+ 1/16) inch above the top pocket opening.	301	SSc-1	6-8	B	B

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TABLE III - CONSTRUCTION OF ALUMINIZED COVER

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	NDL	BOB/ LPR
6.	Bind radio pocket opening. (Cont'd)					
	f. Center a 1 inch by 2 inch piece of pile tape on the inside of the lower radio pocket opening in accordance with marks on patterns. Stitch tape along both sides and top and bottom edges.	301	LSbj-1	6-8	B	B
	g. Postition a 1 inch by 2 inch piece of hook tape on the outside of the lower radio pocket opening as indicated by marks on pattern. Stitch tape along both sides and top and bottom edges.	301	LSbj-1	6-8	B	B
	h. As an alternate, operations 6.f and 6.g may be done at one time.	301	LSbj-1	6-8	B	B
7.	Make and position radio pocket flap.					
	a. Fold radio pocket flap in half face to face, as indicated by marks on patterns and stitch 3/8 inch to 1/2 inch from the raw side edges.	301	SSe-2(a)	6-8	B	B
	b. Turn radio pocket flap to finished position and stitch the turned in sides and bottom folded edge 1/4 inch to 5/16 inch gauge. The corners of the flap shall be completely forced out.	301	SSe-3(b)	6-8	B	B
	c. Position a 1 inch by 2 inch piece of pile tape on the underside of the radio pocket flap as indicated by marks on patterns. Stitch tape along all edges.	301	LSbj-1	6-8	B	B
	d. Turn under 1/4 inch to 3/8 inch the top raw edge of the patch pocket flap and double stitch 1/4 to 5/16 inch gauge to each front as indicated by marks on patterns. The upper row of topstitching shall be 1/8 (± 1/16) inch above top folded edge of the pocket.	301 or 401	LSba-2	6-8	B	B
	or					

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TABLE III - CONSTRUCTION OF ALUMINIZED COVER

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	NDL	BOB/ LPR
7.	Make and position radio pocket flap. (Cont'd)					
	e. As an alternate, position the radio pocket flap face down on coat front, with raw edges facing towards the bottom of the coat and with the raw edges 3/8 inch below the mark on the pattern. Stitch 1/4 to 5/16 inch from raw edge. Turn flap down to finished position and topstitch 3/8 to 7/16 inch from top folded edge of flap.	301 or 401	LSbl-2	6-8	B	B
	f. Finished radio pocket flaps for all sizes shall measure 4-3/4 (± 1/4) inches long when measured along the bottom edge and 3-1/4 (± 1/4) inches wide at side and center.					
	g. Position leather triangular shaped reinforcement pieces across both ends of the patch pocket opening with the base of the triangle parallel with the edges of the pocket flap. Stitch 1/16 to 1/8 inch from all raw edges of the triangle.	301	LSbj-1	6-8	B	B
8.	Assemble aluminized cover.					
	a. Join, face to face, 3/8 to 1/2 inch from raw edge, the fronts and back at shoulder. Turn back panel over seam allowances and topstitch 1/16 to 1/8 inch from folded edge.	301	LSq-2	6-8	B	B
	b. Join face to face 3/8 to 1/2 inch from the raw edge, the sleeves to the armhole, matching notches. Turn back the front and back panels over the seam allowances and topstitch 1/16 to 1/8 inch from folded edge.	301	LSq-2	6-8	B	B

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TABLE III - CONSTRUCTION OF ALUMINIZED COVER

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	NDL	BOB/ LPR
8.	Assemble aluminized cover. (Cont'd)					
	c. Join face to face 3/8 to 1/2 inch from the raw edge, the side of coat and underarm sleeve, simultaneously catching the back edge of the pocket facings, in one continous operation. Turn back the back panel and sleeve over the seam allowance and topstitch 1/16 to 1/8 inch from folded edge.	301	LSq-2	6-8	B	B
	d. Position neck facings face to face with back raw edges even and stitch 3/8 to 1/2 inch from the raw edges. Spread open seam allowance.	301	SSa-1	6-8	B	B
	e. Turn in raw edges of fronts as indicated by the outside fold line on the patterns. Fold front edge face to face along the inside fold line indicated on the pattern.					
	f. Position neck facing on the outershell face to face, matching center back notch with joining seam of neck facing and stitch 1/4 to 3/8 inch from the raw edges.	301	SSa-1	6-8	B	B
	g. Trim ends of neck facing and turn neck edge and fronts to finished position and edgestitch 1/4 (+ 1/16) inch and stitch again 1-1/2 (+ 1/16) inch from the folded neck and 1 (+ 1/16) inch from the front edges.	301	SSae-3 (b)	6-8	B	B
	h. Position three leather reinforcement tabs on the left front 1/16 inch from the front folded edge in accordance with the pattern marker. The points on the tabs shall point away from the front opening and the length of the tabs shall be parallel to the bottom edge of the coat. Double stitch the edges and the point of the reinforcement tab 1/8 to 3/16 inch gauge with the outside row of stitching 1/16 inch from the raw edge.	301	LSbj-2	6-8	B	B

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TABLE III - CONSTRUCTION OF ALUMINIZED COVER

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	NDL	BOB/ LPR
8.	Assemble aluminized cover. (Cont'd)					
	i. Slash leather tab and aluminized coat front opening as indicated on the reinforcement tab marker. Reinforce slashed opening by stitching 1/8 inch around the opening starting at the front folded edge of the coat and continuing around the periphery and stopping at the front folded edge.	301	LSbj-1	6-8	B	B
	j. Turn under entire bottom edge of coat 1/4 to 3/8 inch and edgestitch. Form a 3/4 (± 1/8) inch wide hem, as indicated by notches on pattern. Ends of hem may be folded square or diagonally. Stitch top edge of hem and diagonally folded ends 1/8 to 3/16 inch from folded edge. Square folded ends shall be stitched 1/4 to 5/16 inch from the folded front edge.	301	EFb-1	6-8	B	B
	k. Position a strip of 1 inch wide hook fastener tape down the right and left fronts with the edge 1/8 to 3/16 inch from the folded front edges and the ends finishing 1/4 inch from the folded top and bottom edges. Stitch along all edges of the tape.	301	LSbj-1	6-8	B	B
	l. Position a two inch wide leather strip onto the inside of sleeve opening approximately 7/8 inch in from each sleeve opening. Stitch 1/8 to 3/16 inch from the edge through all plies of material, overlapping ends of the strip 1 to 1-1/2 inch near the underarm seam. Turn strip to the outside of the sleeve opening and stitch 1/8 to 3/16 inch from the edge along free edge and across the overlapped portion through all plies of material.	301	LSa-1 (each edge)	6-8	B	B
	or					

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TABLE III - CONSTRUCTION OF ALUMINIZED COVER

NO.	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH IN	NDL	BOB/ LPR
8.	Assemble aluminized cover. (Cont'd)					
	m. As an option, fold leather strip in half lengthwise and using a folder, stitch each sleeve opening with a single or double row of stitching 1/8 to 3/16 inch from edges of leather strip, through all plies of material, overlapping ends of strip 1 to 1-1/2 inches near the underarm seam. The ends of the sleeves shall extend within and to the folded edge of the leather strip.	301	BSa-1 or BSa-2	6-8	B	B
	n. Attach snap fastener caps and sockets, one each to each shoulder seam 3/4 to 7/8 inch from finished edge of neck opening. Attach a snap fastener cap and socket to bottom hem, mid-way between side seams and centered 1/2 to 5/8 inch from bottom finished edge. Sockets shall be on the inside of the aluminized cover facing the wearer.					
9.	Labels.					
	a. Position and edgestitch the combination label, (see 3.3.13.3) to inside center back of cover 1-1/2 to 2 inches below raw edge of neck facing.	301	LSbj-1	6-8	B	B
	b. Apply warning label, (see 3.3.13.4) horizontally across center back 2 to 3 inches below bottom edge of combination label.					
10.	Clean aluminized covers.					
	a. Trim all threads ends.					
	b. Remove all loose threads.					
	c. Remove all spots and stains.					
	NOTE: Care shall be taken in the use of solvents for cleaning to assure that the aluminized coating is not damaged.					

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3.8 Sizes and measurements. Sizes and measurements of the finished structural coats and the aluminized proximity outer shell shall be as shown in Tables IV and V (see Figures 1 and 2). All measurements and tolerances are expressed in inches.

TABLE IV - STRUCTURAL COAT

SIZE	(A) CHEST	(B) BOTTOM	(C) BACK LENGTH	(D) SLEEVE LENGTH
Small (34-36)	48	50	30	22-1/2
Medium (38-40)	52	54	30-3/4	23
Large (42-44)	56	58	31-1/2	23-1/2
X-Large (46-48)	60	62	32-1/4	24
Tolerance	+1 -1/2	±1	±1/2	±1/2

NOTE: The coat shall be laid flat and smooth with the front dee rings and snaps closed.

(A) Chest - Twice the measurements taken across front from folded edge to folded edge at base of armhole.

(B) Bottom - Twice the measurement taken along bottom edge of coat from folded edge to folded edge.

(C) Back Length - Measure from collar joining seam to bottom of coat along center of back.

(D) Sleeve Length - Measure along underarm seam from base of armhole to bottom edge of sleeve.

TABLE V - ALUMINIZED PROXIMITY COVER

SIZE	(A) CHEST	(B) BOTTOM	(C) BACK LENGTH	(D) SLEEVE LENGTH
Small (34-36)	50	52	32	23-3/4
Medium (38-40)	54	56	32-3/4	24-1/4
Large (42-440	58	60	33-1/2	24-3/4
X-Large (46-48)	62	64	34-1/4	25-1/4
Tolerance	+1 -1/2	±1	±1/2	±1/2

NOTE: The outer cover shall be laid flat and smooth with front edges placed together.

(A) Chest - Twice the measurements taken across front from folded edge to folded edge at base of armhole.

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(B) Bottom - Twice the measurement taken along bottom edge of outer shell from folded edge to folded edge.

(C) Back Length - Measure from neckline to bottom of coat along center of back.

(D) Sleeve Length - Measure along underarm seam from base of armhole to bottom edge of sleeve.

3.9 Workmanship. The finished coats shall conform to the quality established by this specification. The occurence of defects shall not exceed the acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the government to acceptance of defective material.

4.1.2 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3).
2. Quality conformance inspection (see 4.4).

4.3 First article inspection. When required, the first article submitted in accordance with 3.2, shall be inspected as specified in 4.4.2 for compliance with design, construction, workmanship, appearance and dimensional requirements.

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(B) Bottom - Twice the measurement taken along bottom edge of outer shell from folded edge to folded edge.

(C) Back Length - Measure from neckline to bottom of coat along center of back.

(D) Sleeve Length - Measure along underarm seam from base of armhole to bottom edge of sleeve.

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4.4 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-1666, except where otherwise indicated.

4.4.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable procurement documents. A certificate of compliance shall be furnished for paragraphs 3.3.6.1 - identification of silicone treatment, 3.3.7 - corduroy cloth requirements and 3.3.12.2 - identification of metal fasteners. In addition, testing shall be performed on the components and characteristics listed in Table VI. The methods of testing specified in FED-STD-191, wherever applicable, and as listed in Table VI shall be followed. The physical and chemical values specified in Section 3, except where otherwise specified, apply to the average of the determinations made on the sample unit for test purposes as specified in the applicable test methods. All test reports shall contain the individual values utilized in expressing the final results. The lot shall be unacceptable if one or more sample units fail to meet any of the test requirements specified. The sample size shall be in accordance with the following:

LOT SIZE	SAMPLE SIZE
800 or less	2
801 thru 22,000	3
22,001 and over	5

The basis for lot size and sample units for testing shall be as follows:

COMPONENT	LOT SIZE EXPRESSION	SAMPLE UNIT
Laminated cloth	yards	2 yards full width
Film only	yards	1-1/2 yards full width
Leather	square feet	1/2 square foot
Cloth, corduroy	yards	2 yards full width
Snaps and dee rings	gross	1 each

TABLE VI - TESTING OF COMPONENTS

COMPONENT	CHARACTERISTIC	REQUIREMENT PARAGRAPH	TEST METHOD/ PARAGRAPH
Laminated cloth	Moisture vapor transmission	3.3.4	ASTM E96 Procedure B 1/ 2/
	Hydrostatic resistance	3.3.4	5512 1/
	Water permeability after synthetic perspiration	3.3.4	5516 3/ 4/
Leather	Silicone identification	3.3.6.1	5/
	Water absorption	3.3.6.1	8111.1 of FED-STD-311 6/

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TABLE VI - TESTING OF COMPONENTS (Cont'd)

COMPONENT	CHARACTERISTIC	REQUIREMENT PARAGRAPH	TEST METHOD/ PARAGRAPH
Corduroy cloth	All requirements	3.3.7	5/
Snaps hooks and Dee rings	Metal identification	3.3.12.2	5/
	Style	3.3.12.2	Visual
	Dimensions	3.3.12.2	Visual

1/ The film side shall face the water.

2/ The free stream air velocity shall be 550 + 50 ft/min as measured 2 inches above the fabric specimen. The air flow shall be measured at least 2 inches away from any other surface. The test shall be run for 24 hrs. and weight measurements shall be taken at only the start and completion of the test. At the start of the 24 hour test period, the air gap between the water surface and the back of the specimen shall be 3/4 inch. Five specimens shall be tested.

3/ The water permeability shall be measured as specified in Method 5516 of FED-STD-191, except that the face side of the laminated cloth shall contact the water. The hydrostatic head shall be 50 cm and shall be held for 3 minutes. The report shall only include measurement of the appearance of water drops. Leakage is defined as the appearance of water any place within the 4-1/2 inch diameter test area. The test may be performed using any test device which tests the same specimen area at the equivalent pressure using an appropriate size specimen. In cases of dispute, the apparatus described in Method 5516 of FED-STD-191 shall be used.

4/ The synthetic perspiration solution shall be made up in a 500ml beaker by combining 3.0g sodium chloride, 1.0g trypticase soy broth powder, 1.0g normal propyl porpionate, and 0.5g lecithin. Add 500 ml of distilled water, add a magnetic stirring bar, and cover the beaker. Place the beaker on a combination hot plate/magnetic stirrer apparatus. While stirring, heat the solution to 50°C until all ingredients are dissolved. While stirring, cool the solution to 35°C, remove cover, and dispense immediately with a pipet or other suitable measuring device. Dispense 2 ml of perspiration solution at 35°C onto the center of an 8 x 8 inch glass. Place a specimen on the glass plate with the cloth side facing the glass. Dispense an additional 2 ml of the synthetic perspiration solution onto the center of the specimen. Place an 8 x 8 inch glass plate on top of the specimen with a 4 lb weight positioned in the center. Three specimens, 8 inches by 8 inches, shall be cut and exposed to perspiration as indicated above. After 16 hrs, remove the specimen (DO NOT RINSE) and test as specified in 3/.

5/ A certificate of compliance shall be acceptable for this characteristic.

6/ The leather shall be immerced for 30 minutes instead of the specified 24 hours.

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4.4.2 Examination of the end item coat and cover. Examination of the end item shall be in accordance with MIL-STD-1666 for Material Defects and Damages (Sec. I), Components and assembly (Sec. II), Cutting (Sec. III), Cleanness (Sec.IV), Seams and Stitchings (Sec. V), Metal Components (Sec VII), Dimensional examination specified in paragraph 4.3, Inspection levels and acceptable quality levels specified in paragraph 4.4, Examination of packaging requirements specified in 4.5 and in accordance with 4.4.2.1. Defects listed in paragraph 4.4.2.1 shall take precedence over defects listed for similar examinations in MIL-STD-1666.

4.4.2.1 Visual examination. The coats and covers shall be examined for defects in construction, design and workmanship as follows:

NOTE: When an asterisk (*) appears, the defect shall be classified as follows:

MAJOR - When seriously affecting serviceability or appearance.
MINOR - When affecting serviceability or appearance, but not seriously.

CLASSIFICATION OF DEFECTS

DEFECT	CLASSIFICATION MAJOR	MINOR
LEATHER		
a. Loose, spongy or flanky	X	
b. Any open vein	X	
c. Any rupture		X
d. Not soft or pliable.		X
e. Any scar, brittle or bony area, weak or thin spot or abraded area	X	
FRONT CLOSURE		
a. Uneven in length at bottom by 1 inch or more when fastened		X
b. Snaps and dee rings out of alignment causing a noticeable gap or pull on front closure when fastened	*	
c. Dee rings not positioned on securing panel as specified on template by 1/2 inch or more		X
SECURING PANEL		
a. Puckered or pleated noticeably, affecting serviceability		X
b. Panels reversed, i.e. right panel on left side or left panel on right side	X	
c. Folded edge of right panel not parallel with front edge by more than 1/2 inch when measured from top to bottom		X

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DEFECT	CLASSIFICATION MAJOR	MINOR
COLLAR		
a. Topcollar tight or twisted	*	
b. Topcollar excessively full		X
c. Excessive fullness or twist on collar stand		X
d. Collar edge uneven or not properly forced out		X
e Collar points uneven by more than 1/4 inch		X
f. Collar off center by more than 1/2 inch		X
g. Finished collar less than 4-1/8 or more than 4-3/8 inches wide at center back		X
POCKET AND FLAP		
a. Pocket and/or flap out of vertical alignment with front edge by more than 3/4 inch		X
b. Pocket and/or flaps out of horizontal alignment by more than 3/4 inch		X
c. Pocket and/or flap poorly shaped, or not of uniform shape		X
d. Pocket and/or flap set on crookedly, affecting appearance		X
e. Pocket and/or flap measurements outside of specified tolerances		X
f. Excessive fullness or twist on outside of flap		X
g. Bellows top edge of pocket caught during pocket attachment to front		X
SLEEVES		
a. Reversed, i.e., left sleeve in right armhole or right sleeve in left armhole	X	
b. Tight at armhole, causing puckers and pleats on front and back	*	
c. Sleeve tabs on inside not positioned as specified		X
d. Snap fastener sockets missing from sleeve tabs	X	
e. Snap fastener sockets facing towards wrist as worn		X
f. Wristlets finish less then 3-3/4 inches or more 4-1/4 inches		X
g. Bottom of sleeve not leather bound		X
h. Leather binding on sleeve bottom overlapped less than 1 inch or more than 1-1/2 inch.		X
BINDING		
a. Binding not securely caught, pulls away from facing easily		X
b. Binding noticeably puckered or pleated		X
c. Binding tight, will not lie flat		X

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DEFECT	CLASSIFICATION MAJOR	MINOR
HANGER LOOP		
a. Omitted or insecurely attached		X
b. Not centered between ends of collar by more than 3/4 inch		X
c. Length of opening less than 2-3/4 or more than 3-1/4 inches		X
BOTTOM HEM		
a. Outside of coat puckered or twisted by stitching forming hem		X
b. Width less than 7/8 or more than 1-1/8 inches		X
c. Irregular in width by more than 1/4 inch		X
REFLECTIVE TAPE		
a. One section missing		X
b. More than one section missing	X	
c. Edge of tape not parallel with bottom edge of coat by more than 1/2 inch		X
FASTENER TAPE		
a. Any section missing	*	
b. Any section incorrect component, i.e. hook portion instead of pile portion or pile portion instead of hook portion	*	
c. Not edge stitched along all edges when specified		X
LEATHER REINFORCEMENT		
a. Shoulder piece missing	X	
b. Shoulder piece misplaced or not attached as specified		X
c. Buttonhole reinforcement missing	X	
d. Buttonhole reinforcement not attached as specified		X
e. Buttonhole reinforcement piece not slit		X
f. Triangular pocket reinforcement piece missing		X
LABEL		
a. Omitted, one or more		X
b. Not attached as specified		X
c. Inscription or instruction omitted, incomplete, illegible or incorrect		X
d. Position not as specified		X
INSTRUCTION MANUAL		
a. Omitted or not positioned as specified		X

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4.4.4 Palletization examination. An examination shall be made to determine that the palletization complies with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one palletized unit load fully packaged. The lot size shall be the number of palletized unit loads in the end item inspection lot. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5 as specified in MIL-STD-105.

Examine	Defect
Finished dimension	Length, width, or height exceeds specified maximum requirements.
Palletization	Pallet pattern not as specified. Interlocking of loads not as specified. Load not bonded with required straps as specified.
Weight	Exceeds maximum load limits.
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.

5. PACKAGING

5.1 Preservation packaging. Preservation-packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Structural coat. Each structural coat shall be folded as follows:

1. Fasten snaps and dee rings.
2. Place coat flat, front down.
3. Fold each side and sleeve towards center back of coat.
4. Each sleeve shall than be placed parallel to the folded sides of the coat.
5. Fold coat in half by bringing the bottom up toward the collar.

The folded structural coat shall measure approximately 23 inches by 15 inches. Each folded coat shall be individually packaged in a snug-fitting, clear polyethylene film bag having a film thickness of not less than 0.003 inch. The bag may be fabricated from polyethylene film tubing or sheeting. All seams and closures shall be formed by heat sealing. A 1/4 inch diameter hole shall be made at one corner of each bag to allow excess air to escape. As an alternate, the polyethylene bag may be of the tuck in and reverse flap type, in which a heat seal closure and corner hole are not required. The instruction manual must be inserted into the polyethylene bag and positioned so that it is visible.

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5.1.1.2 Aluminized covers. Each aluminized outershell shall be folded as follows:

1. Place cover flat face up.
2. Fold each sleeve over front and down so that top edge of sleeve is parallel to front opening.
3. Fold outershell in half by placing right side over left side.
4. Fold outershell in half again by bringing the bottom up towards the top.

The folded covers shall measure approximately 15 inches by 15 inches. Each folded covers shall be individually packaged in a snug-fitting, clear polyethylene film bag having a film thickness of not less than 0.003 inch. The bag may be fabricated from polyethylene film tubing or sheeting. All seams and closures shall be formed by heat sealing. A 1/4 inch diameter hole shall be made at one corner of each bag to allow excess air to escape. As an alternate, the polytheylene bag may be of the tuck in and reverse flap type, in which a heat seal closure and corner hole are not required. The instruction manual must be inserted into the polyethylene bag and positioned so that it is visible.

5.1.2 Level C (Commercial packaging). Each structural coat and aluminized cover shall be packaged separately to afford adequate protection against physical damage during shipment from the supplier to the first receiving activity. The package and the quantity per package shall be the same as that normally used by the supplier for retail distribution.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Structural coat. Eight structural coats, of one size only, packaged as specified in 5.1, shall be packed in a fiberboard shipping container assembled, closed and reinforced conforming to type CF, class weather- resistant, grade V15c, variety DW, size 3A of MIL-B-17757. The fiberboard used for the liner shall conform to type CF, class domestic conforming to MIL-B-17757. Level A folded coats shall be stacked flat with each coat reversed end to end, eight in depth within the shipping container. The length of the coats shall be parallel to the side panels of the shipping container.

5.2.1.2 Aluminized covers. Twenty covers of one size only, packaged as specified in 5.1, shall be packed in a fiberboard shipping container assembled, closed and reinforced conforming to type CF, class weather-resistant, grade V15c, variety DW, size 3A of MIL-B-17757. The fiberboard used for the liner shall conform to type CF, class domestic conforming to MIL-B-17757. Level A folded covers shall be be stacked on edge within the shipping container with the face of the package parallel to the end panels within the shipping container.

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5.2.2 Level B.

5.2.2.1 Structural coat. Eight structural coats, of one size only, packaged as specified in 5.1, shall be packed in a fiberboard shipping container assembled, closed and reinforced conforming to type CF, class domestic, grade 275, variety DW, size 3A of MIL-B-17757. The fiberboard used for the liner shall conform to type CF, class domestic conforming to MIL-B-17757. Level A folded coats shall be stacked flat with each coat reversed end to end, eight in depth within the shipping container. The length of the coats shall be parallel to the side panels of the shipping container.

5.2.2.2 Aluminized cover. Twenty covers of one size only, packaged as specified in 5.1, shall be packed in a fiberboard shipping container assembled, closed and reinforced conforming to type CF, class domestic, grade 275 variety DW, size 3A of MIL-B-17757. The fiberboard used for the liner shall conform to type CF, class domestic conforming to MIL-B-17757. Level A folded covers shall be be stacked on edge within the shipping container with the face of the package parallel to the end panels within the shipping container.

5.2.3 Level C (Commercial packing). Structural coats and aluminized covers separately packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. The quantity per shipping container shall be the same as that normally used by the supplier for retail distribution. Containers shall comply with U.S Postal Service Manual, Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.3 Palletization. When specified (see 6.2), item, packed as specified in 5.2, shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Each prepared load shall be banded with primary and secondary straps in accordance with the banding means C, K and L, or O or P. Pallet patterns shall be in accordance with the appendix of MIL-STD-147. The pallet shall be 4-way, Type IV; Type V, class 1, size 2; or Type VIII, fabricated from wood groups I, II, III or IV, Grade A of NN-P-71, or 4-way, Style 1, Size A, Type I, Class 1 fabricated from groups specified of MIL-P-15011. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contacting officer.

5.4 Marking. In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

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5.4.1 Polyethylene bagged packages. Polyethylene bagged packages shall have the required identification information legibly printed or stamped in black directly on the bag across the center face or on a white paper label inserted within the bag so as to permit ready identification. The bag or label shall only indicate the following information:

STOCK NUMBER
NOMENCLATURE
SIZE
QUANTITY

6. NOTES

6.1 Intended use. The coats are intended for wear by DOD personnel engaged in firefighting. The structural coat is for use by firefighters engaged in structural type fires. The structural coat with the aluminized outer shell attached is for use by firefighters engaged in proximity type fire fighting.

6.2 Ordering data. Procurement documents should specify the following:

a. Title and date of this specification.
b. Types, classes and sizes required (see 1.2)
c. When first article inspection is required (see 3.2), the item will be tested and should be a first article sample. The contracting officer should include specific instructions in acquistion documents regarding arrangements for examination, quantity and testing and approval of the first article.
d. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
e. List of approved silicone compounds (see 3.3.6.1).
f. When palletization is required (see 5.3).

6.3 Samples and patterns. For access to samples and patterns, address the procuring activity issuing the invitation for bids.

6.4 Recycled materials. It is encouraged that recycled materials be used when practical as long as it meets the requirements of this specification.

6.5 A suggested source of supply for the laminated material specfied in 3.3.4 is W. L. Gore and Associates, Inc., 3 Blue Ball Rd., P.O. Box 1130, Elkton, Maryland 21921.

6.6 A suggested source of supply for the reflective tape specified in 3.3.9 is 3M Safety and Security Systems Division, St. Paul, Minnesota 55144.

6.7 A suggested source of supply for the plated zinc snap hooks and dee rings specified in 3.3.12.2 is Custom Metal Crafters Inc., Newington, CT 06111.

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6.8 Silicone compounds and flame resistant treatments. Only those silicone compounds and flame resistant treatments approved by the U.S. Army Natick Research, Development and Engineering Center, Natick, Massachusetts 01760, and the appropriate medical service and so listed in the invitation for bids or request for proposal, shall be considered acceptable for the related procurement (see 3.3.6.1 and 3.3.7).

6.9 Equal item. Prior to the use of an "or equal" item, the contractor shall submit the item with supporting data to the contracting officer for subsequent approval or disapproval by the responsible Military Agency.

6.10 Subject term (key word) listing.
Cloth, para-aramid/PBI
Cloth, aluminized
Clothing, protective
Coat, combination
Coat, proximity
Coat, structural

6.11 Liner. The liner mentioned in the warning labels specified in paragraphs 3.3.13.1 and 3.3.13.2 refers to the liner covered by specification; MIL-L-24943 - Liner, Fireman's Coat, Quilted.

Custodians:
Navy - NU
Army - GL
Air Force - 99

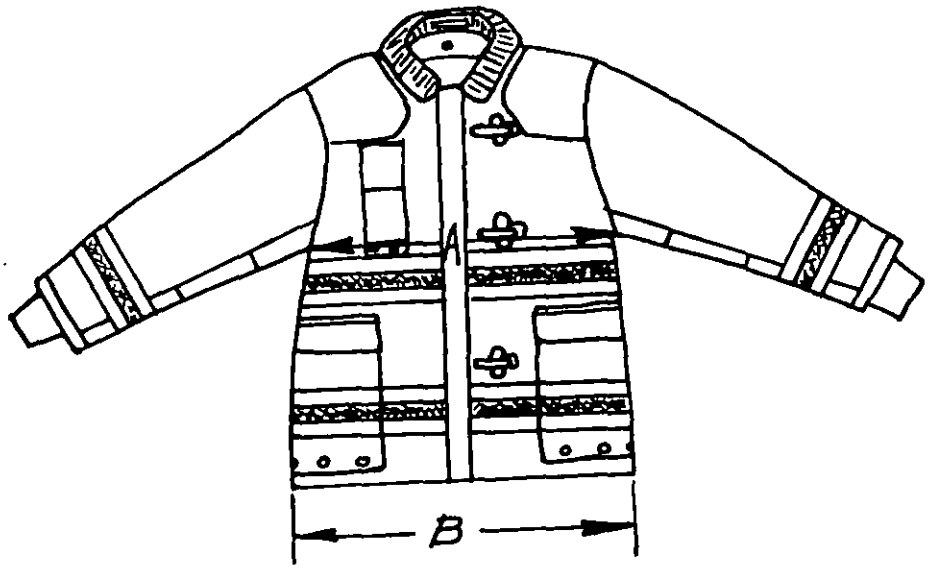
Review activities:
Navy - MC
Army - MD
Air Force - 50, 82
DLA - CT

User activities:
Navy - SH, CG
Air Force - 45

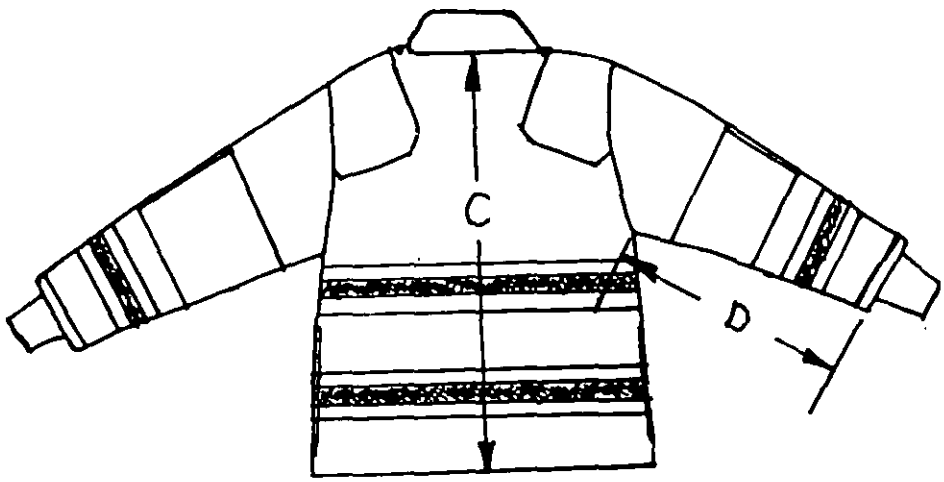
Preparing activity:
Navy - NU

Project No. 8415-0629

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FRONT VIEW

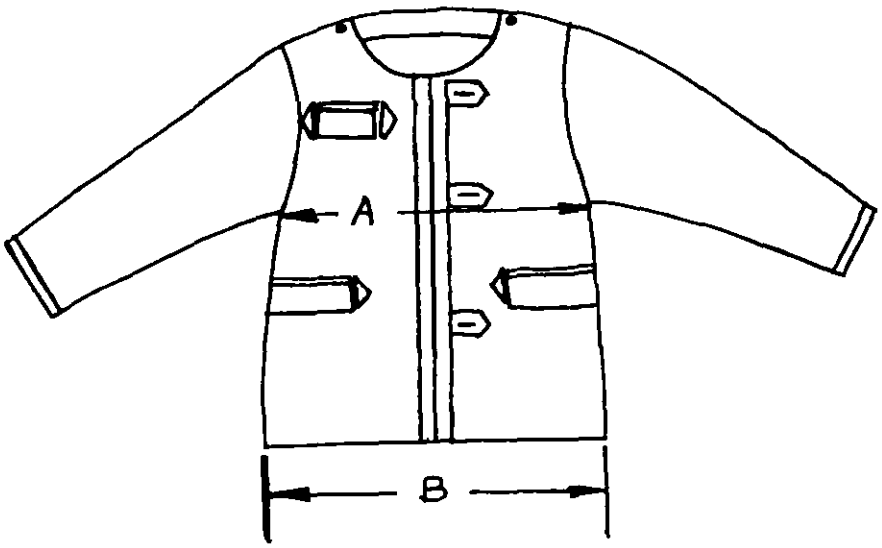


BACK VIEW

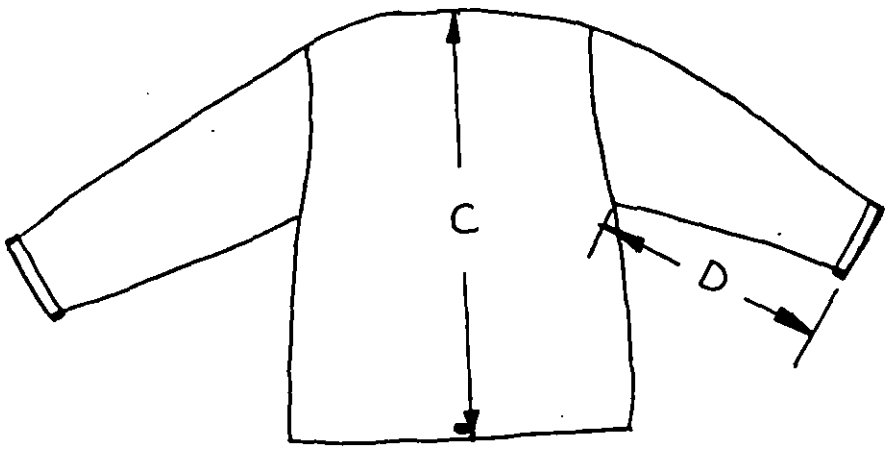
FIGURE 1

FIREMEN'S STRUCTUAL COAT

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FRONT VIEW



BACK VIEW

FIGURE 2

ALUMINIZED PROXIMITY OUTERSHELL

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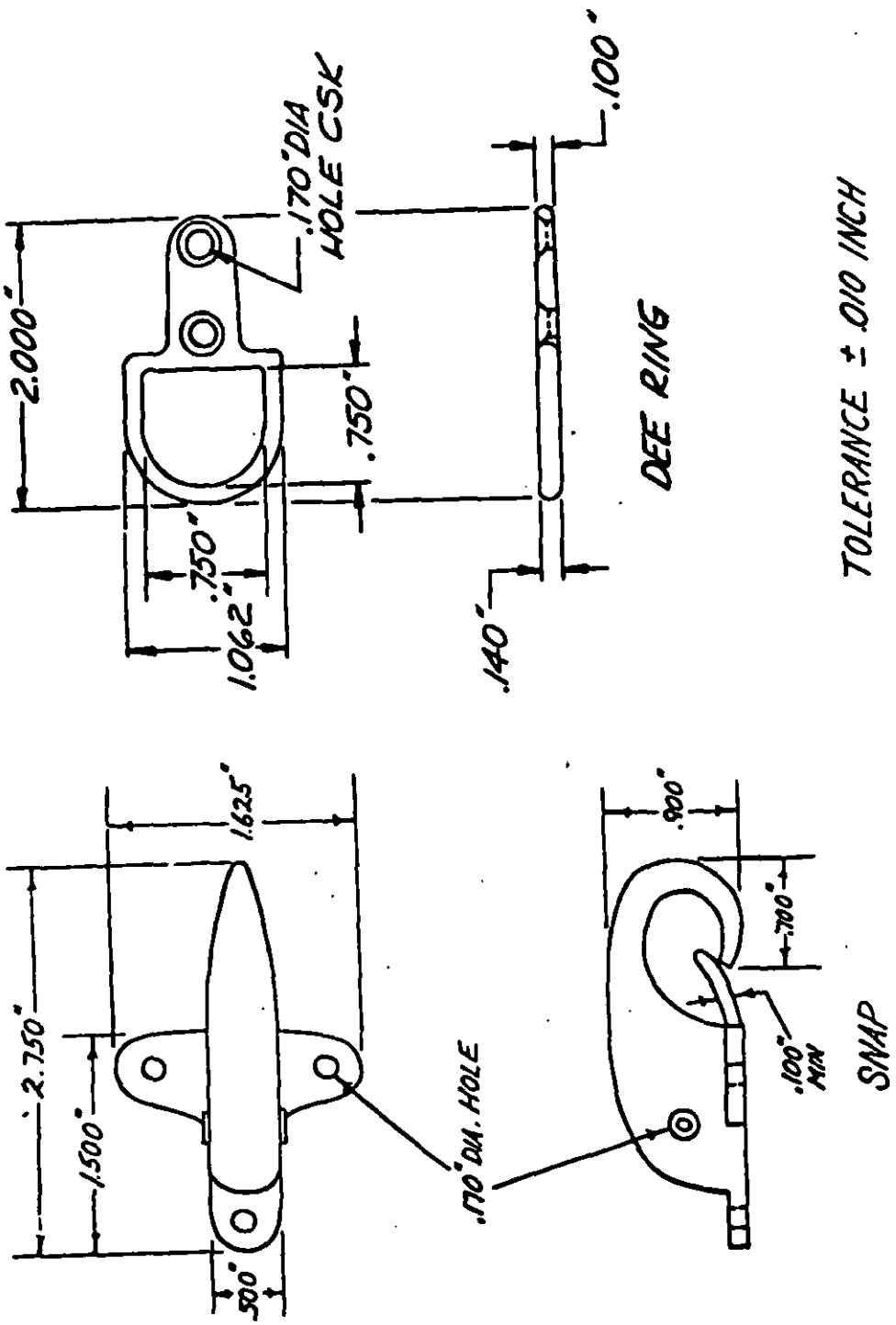


FIGURE 3

SNAP HOOK AND DEE RING

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-24938	COATS, FIREMEN'S, STRUCTURAL/ALUMINIZED PROXIMITY

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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